

Practice Worksheet 3

Section 1: Multiple Choice Questions

(No calculator is allowed for these questions. Choose the best answer.)

1. What is the exact value of $\tan\left(\frac{5\pi}{6}\right)$?

(A) $\sqrt{3}$

(B) $\frac{\sqrt{3}}{3}$

(C) $-\frac{\sqrt{3}}{3}$

(D) $-\sqrt{3}$

2. Evaluate exactly: $\arcsin\left(\sin\left(\frac{7\pi}{6}\right)\right)$.

(A) $\frac{7\pi}{6}$

(B) $\frac{5\pi}{6}$

(C) $\frac{\pi}{6}$

(D) $-\frac{\pi}{6}$

3. How many solutions does the equation $\tan^2 x - 3 = 0$ have in the interval $[0, \pi]$?

(A) 1

(B) 2

(C) 3

(D) 4

4. Find the exact value of $\cos\left(\arctan\left(-\frac{3}{4}\right)\right)$.

(A) $-\frac{4}{5}$

(B) $-\frac{3}{5}$

(C) $\frac{3}{5}$

(D) $\frac{4}{5}$

5. Which of the following represents all solutions to the equation $2\cos^2 x + \cos x - 1 = 0$ on the interval $[0, 2\pi)$?

(A) $x = \frac{\pi}{3}, \frac{5\pi}{3}$

(B) $x = \frac{\pi}{3}, \pi, \frac{5\pi}{3}$

(C) $x = \frac{\pi}{6}, \pi, \frac{11\pi}{6}$

(D) $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$

6. Which of the following statements about the function $f(x) = \tan x$ is FALSE?

(A) The period of $f(x)$ is π .

(B) The range of $f(x)$ is $(-\infty, \infty)$.

(C) $f(x)$ is strictly increasing on the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$.

(D) $f(x)$ has a vertical asymptote at $x = \pi$.

7. What is the domain of the function $g(x) = \arcsin(2x - 1)$?

(A) $[-1, 1]$

(B) $[0, 1]$

(C) $[-\frac{\pi}{2}, \frac{\pi}{2}]$

(D) $(-\infty, \infty)$

8. On the interval $[0, 2\pi)$, what is the solution set for the inequality $\tan x \geq 1$?

(A) $[\frac{\pi}{4}, \frac{\pi}{2}) \cup [\frac{5\pi}{4}, \frac{3\pi}{2})$

(B) $[\frac{\pi}{4}, \frac{5\pi}{4}]$

(C) $[\frac{\pi}{4}, \frac{\pi}{2}] \cup [\frac{5\pi}{4}, \frac{3\pi}{2}]$

(D) $[\frac{\pi}{4}, \infty)$

Section 2: Free Response Questions

(Show all your work clearly.)

FRQ 1 Let $f(x) = \arctan(x)$ and $g(x) = \tan(x)$.

(a) Evaluate $f(\sqrt{3})$ and $f(-\sqrt{3})$. State the range of $f(x)$.

(b) Is the equation $g(f(x)) = x$ true for all real numbers x ? Justify your reasoning based on the domains and ranges of the functions.

(c) Is the equation $f(g(x)) = x$ true for all x in the domain of g ? If yes, justify. If no, provide a specific counterexample by evaluating both sides of the equation.

FRQ 2 Consider the function $h(x) = 2 \sin^2 x - \sin x$.

- (a) Find all zeros of $h(x)$ in the restricted interval $[0, 2\pi)$.
- (b) Solve the inequality $h(x) > 0$ on the interval $[0, 2\pi)$. Present your answer using interval notation.
- (c) If the domain is extended to $(-\infty, \infty)$, write the general solution for the equation $h(x) = 1$.

FRQ 3 Let $k(x) = \arcsin(2x) + \frac{\pi}{6}$.

- (a) Determine the domain and range of $k(x)$.
- (b) Solve the equation $k(x) = \frac{\pi}{2}$. Show the steps of your algebraic manipulation.
- (c) Solve the inequality $k(x) < 0$. Be sure to consider the domain of the function in your final answer.